Predictive Modeling with Regression

Concept of Regression Analysis

- **Regression analysis** is a statistical method used to **examine the relationship** between a **dependent variable** (the outcome) and one or more **independent variables** (predictors).
- Its main goal is to **predict** the value of the dependent variable and **understand how changes** in the independent variables affect it.
- **Example**: Predicting a student's score based on study hours and attendance.

Purpose:

- To **predict outcomes** (e.g., sales, prices, marks, growth rate, etc.)
- To **quantify relationships** (e.g., how much change in X affects Y)

Building Linear Regression Models

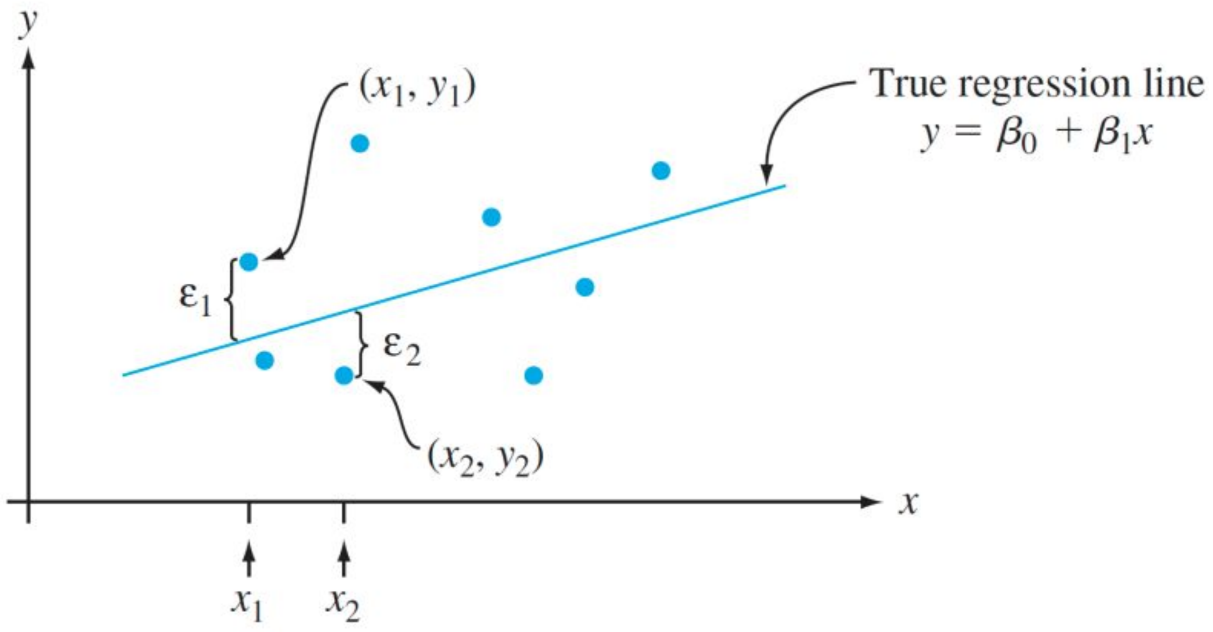
- **Linear regression** models assume a **linear relationship** between the dependent variable Y and independent variable(s) X.
- **Equation:**

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where

- β_0 = intercept
- β_1 = slope (how much Y changes with one unit change in X)
- ϵ = error term

- Steps to build a model:
  1. **Collect data** for dependent and independent variables.
  2. **Split data** into training and testing sets.
  3. **Fit the model** using training data (find the best-fit line).
  4. **Evaluate performance** using metrics like R^2 and RMSE.
  5. **Use the model** to make predictions.



Interpretation of the Best-Fit Line

- **Slope (m):** The slope of the best-fit line indicates how much the dependent variable (y) changes with each unit change in the independent variable (x). For example if the slope is 5, it means that for every 1-unit increase in x, the value of y increases by 5 units.

- **Intercept (b):** The intercept represents the predicted value of y when x = 0. It’s the point where the line crosses the y-axis.

Minimizing the Error: The Least Squares Method

To find the best-fit line, we use a method called **Least Squares**.

The idea behind this method is to minimize the sum o: squared differences between the actual values (data points) and the predicted values from the line.

These differences are called residuals.

The formula for residuals is:

$$Residual = y_i - \hat{y}_i$$

Where:

- y_i is the actual observed value
- $\hat{y}_i$ is the predicted value from the line for that x_i

The least squares method minimizes the sum of the squared residuals:

$$Sum of squared errors (SSE) = \Sigma(y_i - \hat{y}_i)^2$$

This method ensures that the line best represents the data where the sum of the squared differences between the predicted values and actual values is as small as possible.

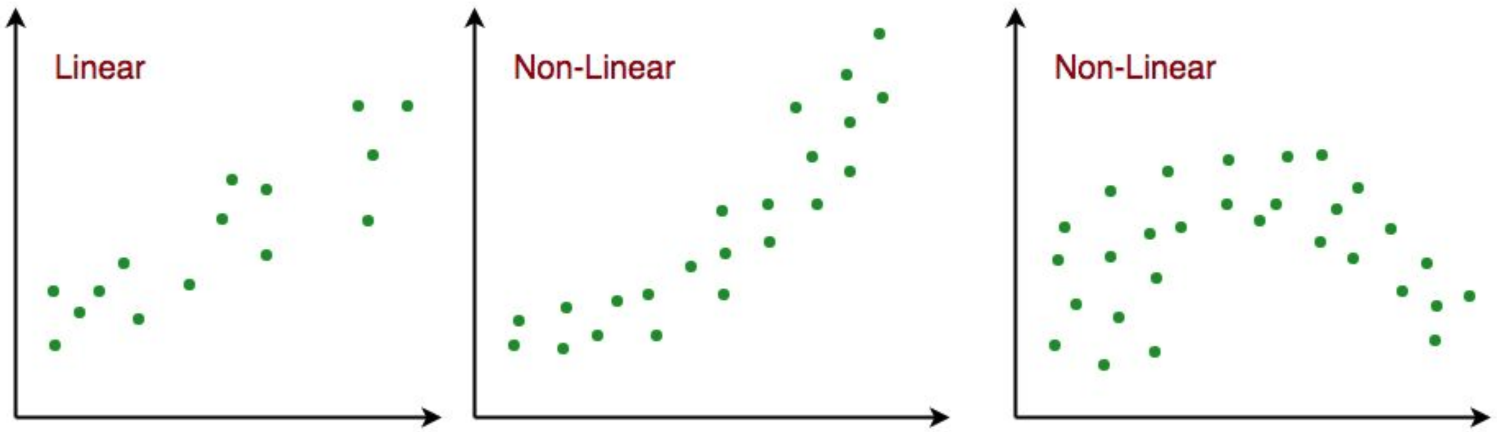
Limitations

- ***Assumes Linearity:*** *The method assumes the relationship between the variables is linear. If the relationship is non-linear, linear regression might not work well.*

- ***Sensitivity to Outliers:*** *Outliers can significantly affect the slope and intercept, skewing the best-fit line.*

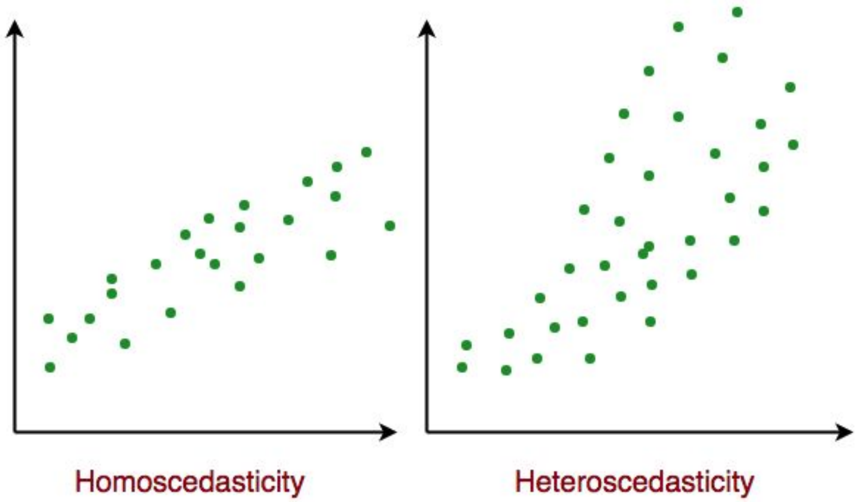
Assumptions of the Linear Regression

1. Linearity: The relationship between inputs (X) and the output (Y) is a straight line.



2. Independence of Errors: The errors in predictions should not affect each other.

3. Constant Variance (Homoscedasticity): The errors should have equal spread across all values of the input. If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.



4. Normality of Errors: The errors should follow a normal (bell-shaped) distribution.

5. No Multicollinearity(for multiple regression): Input variables shouldn't be too closely related to each other. **Multicollinearity** refers to a situation in **multiple regression analysis** where **two or more independent (predictor) variables are highly correlated** with each other.

6. No Autocorrelation: Errors shouldn't show repeating patterns, especially in time-based data.

Use Case of Multiple Linear Regression

Multiple linear regression allows us to analyze relationship between multiple independent variables and a single dependent variable. Here are some use cases:

- **Real Estate Pricing:** In real estate MLR is used to predict property prices based on multiple factors such as location, size, number of bedrooms, etc. This helps buyers and sellers understand market trends and set competitive prices.

- **Financial Forecasting:** Financial analysts use MLR to predict stock prices or economic indicators based on multiple influencing factors such as interest rates, inflation rates and market trends. This enables better investment strategies and risk management24.

- **Agricultural Yield Prediction:** Farmers can use MLR to estimate crop yields based on several variables like rainfall, temperature, soil quality and fertilizer usage. This information helps in planning agricultural practices for optimal productivity

- **E-commerce Sales Analysis:** An e-commerce company can utilize MLR to assess how various factors such as product price, marketing promotions and seasonal trends impact sales.

Use Case Example

- **Business:** Predict future sales based on advertising spend.
- **Healthcare:** Estimate blood pressure based on age, weight, and activity level.
- **Education:** Predict student performance from attendance and assignment scores.

Where Not to Use

- When the relationship between variables is **non-linear** (use nonlinear or tree-based models).
- When data contains **many outliers** or **strong multicollinearity** among predictors.
- When predictors are **categorical without proper encoding**.

Tips

- Always **visualize data** first (scatterplots help identify trends).
- Use **correlation** to detect multicollinearity.
- **Normalize or standardize** variables if scales differ widely.
- Split data into **training and testing sets** to avoid overfitting.

Evaluation Metrics for Linear Regression

1. Mean Square Error (MSE)

Mean Squared Error (MSE) is an evaluation metric that calculates the average of the squared differences between the actual and predicted values for all the data points. The difference is squared to ensure that negative and positive differences don't cancel each other out.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y_i}\right)^2$$

Here,

- n is the number of data points.
- y_i is the actual or observed value for the i^{th} data point.
- $\widehat{y_i}$ is the predicted value for the i^{th} data point.

MSE is a way to quantify the accuracy of a model's predictions. MSE is sensitive to outliers.

2. Mean Absolute Error (MAE)

Mean Absolute Error is an evaluation metric used to calculate the accuracy of a regression model. MAE measures the average absolute difference between the predicted values and actual values.

Mathematically MAE is expressed as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \widehat{Y_i}|$$

Here,

- n is the number of observations
- Y_i represents the actual values.
- $\widehat{Y_i}$ represents the predicted values

Lower MAE value indicates better model performance. It is not sensitive to the outliers as we consider absolute differences.

3. Root Mean Squared Error (RMSE)

The square root of the residuals' variance is the Root Mean Squared Error. It describes how well the observed data points match the expected values or the model's absolute fit to the data. In mathematical notation, it can be expressed as:

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

RSME is not as good of a metric as R-squared. Root Mean Squared Error can fluctuate when the units of the variables vary since its value is dependent on the variables' units (it is not a normalized measure).

4. Coefficient of Determination (R^2)

The **R-square (R^2)**, also called the **coefficient of determination**, is a **statistical measure** that shows how well the independent variables explain the variability of the dependent variable in a regression model.

- **Range:** 0 → 1

Interpretation

R^2 = 0 → Model explains **none** of the variation in the dependent variable.

R^2 = 1 → Model explains **all** of the variation perfectly.

Higher R^2 means the model fits the data better (but not necessarily that it's the best model).

◆ **Definition**

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

where:

- SS_{res} = Sum of Squares of Residuals = $\sum(y_i - \hat{y}_i)^2$
- SS_{tot} = Total Sum of Squares = $\sum(y_i - \bar{y})^2$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

where

- y_i = each observed value of the dependent variable
- n = total number of observations

$$SS_{tot} = \sum (y_i - \bar{y})^2$$

It measures how much the data points vary **around their mean** — before any regression model is applied.

Example

If you build a regression model to predict **student marks (Y)** using **hours studied (X)** and you get:

$$R^2 = 0.85$$

That means **85% of the variation in marks** is explained by the number of hours studied, and **15% is due to other unexplained factors**.

(b) Adjusted R-squared

- **Definition:** Modified version of R^2 that **penalizes unnecessary predictors**.
- **Use:** Better for comparing models with different numbers of predictors.
- **Rule:** If Adjusted $R^2 \approx R^2 \rightarrow$ predictors are meaningful.

If Adjusted $R^2 \ll R^2 \rightarrow$ model may be overfitting.

Adjusted R-square helps to prevent overfitting. It penalizes the model with additional predictors that do not contribute significantly to explain the variance in the dependent variable.

Numerical

We have 6 observations:

x (predictor): 1, 2, 3, 4, 5, 6

y (response) : 2, 3, 5, 4, 6, 7

We'll fit the model:

$$\hat{y} = b_0 + b_1 x$$

Step 1 — basic statistics

- $n = 6$
- $\bar{x} = \dfrac{1+2+3+4+5+6}{6} = 3.5$
- $\bar{y} = \dfrac{2+3+5+4+6+7}{6} = 4.5$

Step 2 — estimate slope b_1 and intercept b_0

Formula for slope:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Plugging numbers (computed exactly):

$$b_1 = 0.942857142857\ldots$$

Intercept:

$$b_0 = \bar{y} - b_1\bar{x} = 1.20$$

So the estimated regression line is:

$$\boxed{\hat{y} = 1.2000 + 0.9428571\,x}$$

Step 3 — predictions ($\hat{y}$) and residuals

Calculate $\hat{y}_i = b_0 + b_1 x_i$, residual $e_i = y_i - \hat{y}_i$:

x	y	ŷ (yhat)	residual = y - ŷ	residual²
1	2	2.142857142857	-0.142857142857	0.02040816327
2	3	3.085714285714	-0.085714285714	0.007346938776
3	5	4.028571428571	0.971428571429	0.943673469388
4	4	4.971428571429	-0.971428571429	0.943673469388
5	6	5.914285714286	0.085714285714	0.007346938776
6	7	6.857142857143	0.142857142857	0.020408163265

Step 4 — sums of squares and R²

- Residual Sum of Squares (SSE) = $\sum(y_i - \hat{y}_i)^2 = 1.942857142857$
- Regression Sum of Squares (SSR, explained) = $\sum(\hat{y}_i - \bar{y})^2 = 15.557142857143$
- Total Sum of Squares (SST) = $\sum(y_i - \bar{y})^2 = 17.5$

Check: $SSE + SSR = 1.942857 + 15.5571429 = 17.5 = SST$ ✅

Coefficient of determination:

$$R^2 = \frac{SSR}{SST} = 0.888979591837 \approx 0.889$$

So about **88.90%** of the variation in y is explained by x.

Step 5 — estimate variance, standard errors, t-tests

Estimated variance of residuals:

$$\hat{\sigma}^2 = s^2 = \frac{SSE}{n-2} = \frac{1.942857142857}{4} = 0.485714285714$$

Standard error of slope b_1:

$$SE(b_1) = \sqrt{\frac{s^2}{\sum(x_i - \bar{x})^2}} = 0.166598625567$$

Standard error of intercept b_0:

$$SE(b_0) = \sqrt{s^2\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right)} = 0.648808431629$$

t-statistics:

- $t_{b_1} = \frac{b_1}{SE(b_1)} = 5.659453$ → p-value ≈ **0.004805** (significant at usual levels)
- $t_{b_0} = \frac{b_0}{SE(b_0)} = 1.849544$ → p-value ≈ **0.138062** (not significant at 0.05)

Degrees of freedom for these tests: $n - 2 = 4$.

Quick interpretation

- Slope $b_1 \approx 0.943$: for each 1-unit increase in x, y increases on average by ≈ 0.943.
- Intercept $b_0 = 1.20$: predicted y when $x = 0$ is 1.20 (interpret with caution if $x = 0$ is outside data range).
- Model explains ≈ 88.9% of the variance — a good fit here.
- Slope is statistically significant (p ≈ 0.0048), intercept is not (p ≈ 0.138).

Logistic Regression for Classification Tasks

Logistic regression is used when the **dependent variable is categorical**, usually **binary** (e.g., Yes/No, 0/1).

It predicts the **probability** that an observation belongs to a particular category.

The output is passed through a **logistic (sigmoid) function** to keep predictions between 0 and 1.

Equation:

$$P(Y=1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Example: Predicting whether an email is *spam (1)* or *not spam (0)*.

Common evaluation metrics: **Accuracy, Precision, Recall, F1-score, ROC–AUC.**

Assumptions of Logistic Regression

Understanding the assumptions behind logistic regression is important to ensure the model is applied correctly, main assumptions are:

1. **Independent observations**: Each data point is assumed to be independent of the others means there should be no correlation or dependence between the input samples.

2. **Binary dependent variables**: It takes the assumption that the dependent variable must be binary, means it can take only two values. For more than two categories SoftMax functions are used.

3. **Linearity relationship between independent variables and log odds**: The model assumes a linear relationship between the independent variables and the log odds of the dependent variable which means the predictors affect the log odds in a linear way.

4. **No outliers**: The dataset should not contain extreme outliers as they can distort the estimation of the logistic regression coefficients.

5. **Large sample size**: It requires a sufficiently large sample size to produce reliable and stable results.

Understanding Sigmoid Function

1. The sigmoid function is an important part of logistic regression which is used to convert the raw output of the model into a probability value between 0 and 1.

2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.

3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.

- If the sigmoid output is same or above the threshold, the input is classified as Class 1.
- If it is below the threshold, the input is classified as Class 0.

This approach helps to transform continuous input values into meaningful class predictions.

Terminologies involved in Logistic Regression

Here are some common terms involved in logistic regression:

1. **Independent Variables:** These are the input features or predictor variables used to make predictions about the dependent variable.

2. **Dependent Variable**: This is the target variable that we aim to predict. In logistic regression, the dependent variable is **categorical**.

3. **Logistic Function**: This function transforms the independent variables into a probability between 0 and 1 which represents the likelihood that the dependent variable is either 0 or 1.

4. **Odds**: This is the ratio of the probability of an event happening to the probability of it not happening. It differs from probability because probability is the ratio of occurrences to total possibilities.

$$\text{Odds} = \frac{p}{1-p}$$

It represents how likely the event is to occur versus not occur.

5.Log-Odds (Logit): The natural logarithm of the odds. In logistic regression, the log-odds are modeled as a linear combination of the independent variables and the intercept.

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$$

This is called the **logit transformation**.

6.Coefficient: These are the parameters estimated by the logistic regression model which shows how strongly the independent variables affect the dependent variable.

7.Intercept: The constant term in the logistic regression model which represents the log-odds when all independent variables are equal to zero.

8.Maximum Likelihood Estimation (MLE): This method is used to estimate the coefficients of the logistic regression model by maximizing the likelihood of observing the given data.

The Logistic (Sigmoid) Function

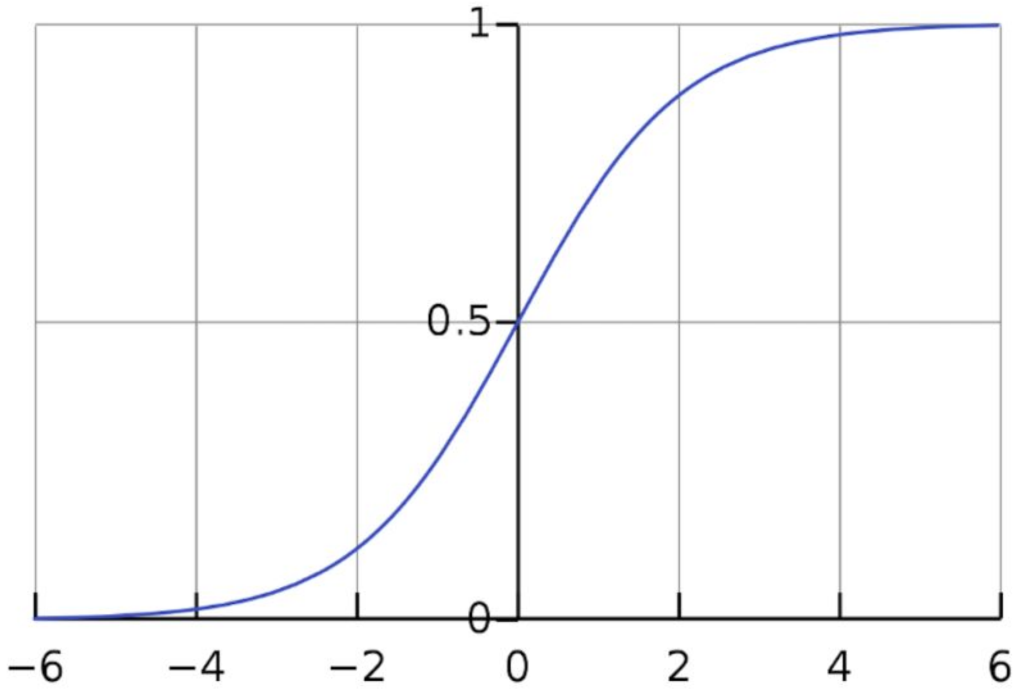
To convert the linear output back to probability:

$$p = \frac{1}{1+e^{-z}}, \quad \text{where } z = b_0 + b_1 x_1 + \cdots + b_n x_n$$

📊 Behavior:

- When $z \to \infty, p \to 1$
- When $z \to -\infty, p \to 0$
- When $z = 0, p = 0.5$

This gives an **S-shaped curve**, perfect for probability modeling.



Decision Boundary

The decision rule in binary classification:

$$\hat{y} = \begin{cases} 1 & if p \geq 0.5 \\ 0 & if p < 0.5 \end{cases}$$

But the threshold (0.5) can be changed (e.g., 0.7) depending on the sensitivity required.

The boundary where $p = 0.5$(or $z = 0$) is the **decision boundary** — often linear if we have two variables.

🧮 Mathematical Representation

1. Start with a linear combination of inputs:

$$z = b_0 + b_1x_1 + b_2x_2 + \ldots + b_nx_n$$

2. Apply the **Sigmoid function** to get a probability:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

3. Decision rule (binary classification):

$$\text{If } P(Y = 1) \geq 0.5, \text{ predict } Y = 1; \text{ else } Y = 0$$

📈 Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- When z is large → output ≈ 1
- When z is small → output ≈ 0

Hours (x)	Pass (y)
1	0
2	0
3	0
4	1
5	1

Assume we fit a model:

$$z = -4 + 1.2x$$

Now, the probability:

$$p = \frac{1}{1 + e^{-(-4+1.2x)}}$$

x	z	p = 1/(1+e^-z)	Prediction
1	-2.8	0.057	Fail (0)
2	-1.6	0.168	Fail (0)
3	-0.4	0.401	Fail (0)
4	0.8	0.690	Pass (1)
5	2.0	0.881	Pass (1)

✅ Hence, as study hours increase, the probability of passing increases.

Logistic Regression Equation and Odds:

It models the odds of the dependent event occurring which is the ratio of the probability of the event to the probability of it not occurring:

$$\frac{p(x)}{1-p(x)} = e^z$$

Taking the natural logarithm of the odds gives the log-odds or logit:

$$\begin{aligned}
\log\left[\frac{p(x)}{1-p(x)}\right] &= z \\
\log\left[\frac{p(x)}{1-p(x)}\right] &= w \cdot X + b \\
\frac{p(x)}{1-p(x)} &= e^{w \cdot X+b} \quad \cdots \text{Exponentiate both sides} \\
p(x) &= e^{w \cdot X+b} \cdot (1 - p(x)) \\
p(x) &= e^{w \cdot X+b} - e^{w \cdot X+b} \cdot p(x)) \\
p(x) + e^{w \cdot X+b} \cdot p(x)) &= e^{w \cdot X+b} \\
p(x)(1 + e^{w \cdot X+b}) &= e^{w \cdot X+b} \\
p(x) &= \frac{e^{w \cdot X+b}}{1 + e^{w \cdot X+b}}
\end{aligned}$$

then the final logistic regression equation will be:

$$p(X; b, w) = \frac{e^{w \cdot X+b}}{1+e^{w \cdot X+b}} = \frac{1}{1+e^{-w \cdot X+b}}$$

This formula represents the probability of the input belonging to Class 1.

🔷 Cost Function

Linear regression uses **Mean Squared Error**, but that doesn't work well for classification.

Logistic regression uses the **Log Loss (Cross-Entropy Loss)**:

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

Where:

- y_i: actual label (0 or 1)
- $\hat{y}_i$: predicted probability

Applications

- Disease detection (e.g., cancer, diabetes)
- Email spam detection
- Customer churn prediction
- Credit risk scoring

Model Evaluation Metrics

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Metric	Formula	Interpretation
Accuracy	(TP + TN) / Total	Overall correctness
Precision	TP / (TP + FP)	Correct positive predictions
Recall (Sensitivity)	TP / (TP + FN)	Ability to detect actual positives
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Balance between precision and recall

Model Evaluation Metrics

ROC Curve	Plot of TPR vs FPR	Visual performance measure
AUC	Area under ROC curve	Higher = better

- **True Positive Rate (TPR)** = Recall = Sensitivity

$$TPR = \frac{TP}{TP + FN}$$

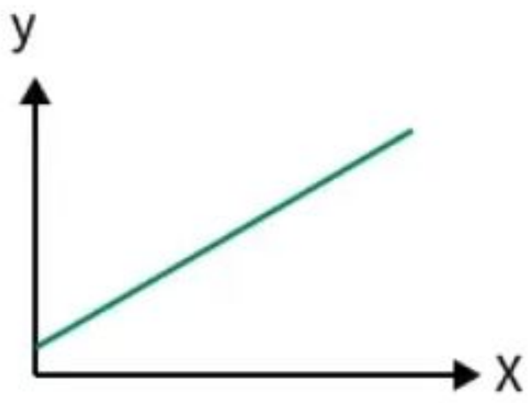
- **False Positive Rate (FPR)**

$$FPR = \frac{FP}{FP + TN}$$

Linear Regression vs Logistic Regression

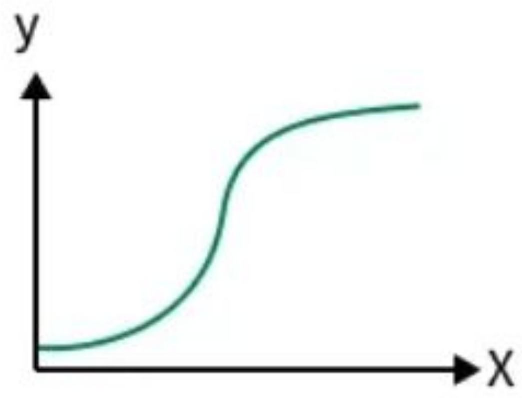
Linear Regression

- Predicts continuous values
- Uses best-fit line
- Solves regression problems



Logistic Regression

- Predicts categorical classes
- Uses sigmoid S-curve
- Solves classification problems



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